

Quick start for LaTeXing with IEEEtran.cls for IEEE Computer Society Conferences

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Abstract—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

I. INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus enim. Vestibulum pellentesque felis eu massa. **TODO!**

The remainder of the paper starts with a presentation of related work (Section II). It is followed by a presentation of hints on LaTeX (Section III). The paper closes with a conclusion and an outlook on future work (Section IV).

II. RELATED WORK

Winery [1] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [2].

To search for related work, ScienceOS¹ can be used; it stores data in the EU and backs its answers with citations, which you still have to read and check yourself. Writing hints for student theses are given in the open-access book *Student Thesis Projects*². More hints are given in *Writing for Computer Science*³.

III. LATEX HINTS

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

A. Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines.

¹<https://www.scienceos.ai/>

²<https://vdf.ch/product/student-thesis-projects-en.html>

³<https://doi.org/10.1007/978-1-4471-6639-9>

To start a new paragraph, press enter twice. This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (`\`).

Running text seldom needs a hard line break. Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. Andrew Stacey gives a long motivation for that in the section “Line Wrapping” of his *Document Revision System*⁴.

Corresponding LaTeX code of `./paper-minted.tex`

```
578 One sentence per line.
579 This rule is important for the usage of version
    ↪ control systems.
580 A new line is generated with a blank line.
581 As you would do in Word:
582 New paragraphs are generated by pressing enter.
583 In LaTeX, this does not lead to a new paragraph
    ↪ as LaTeX joins subsequent lines.
584 To start a new paragraph, press enter twice.
585 This leads to an empty line.
586 In word, there is the functionality to press
    ↪ shift and enter.
587 This leads to a hard line break.
588 The text starts at the beginning of a new line.
589 In LaTeX, you can do that by using two
    ↪ backslashes (\textbackslash\textbackslash).
590 \
591 Running text seldom needs a hard line break.
592
593 Please do \textit{not} use two backslashes for
    ↪ new paragraphs.
594 For instance, this sentence belongs to the same
    ↪ paragraph, whereas the last one started a
    ↪ new one.
595 Andrew Stacey gives a long motivation for that
    ↪ in the section \enquote{Line Wrapping} of
    ↪ his \href{https://loopspace.mathforge.org/}
    ↪ HowDidIDoThat/TeX/VCS/#section.3}{\emph{
    ↪ Document Revision
    ↪ System}}\footnote{\url{https://
    ↪ loopspace.mathforge.org/HowDidIDoThat/TeX/
    ↪ VCS/}}.
```

B. Notes separated from the text

The package `mindflow` enables writing down notes and annotations in a way so that they are separated from the main text.

This is a small note.

Corresponding LaTeX code of `./paper-minted.tex`

```
603 \begin{mindflow}
604 This is a small note.
605 \end{mindflow}
```

⁴<https://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/>

C. Handling TODOs

Marked text.

Corresponding LaTeX code of `./paper-minted.tex`

```
611 \textmarker{Marked text.}
```

With `\textmarker`, only the text color is changed, since this also works well for individual words.

Marked text.

Corresponding LaTeX code of `./paper-minted.tex`

```
617 \textcomment{Marked text.}{A comment on it.}
```

Manual marking for text that has been changed since the last version.

Corresponding LaTeX code of `./paper-minted.tex`

```
621 \modified{Manual marking for text that has been
    ↪ changed since the last version.}
```

This is some text. Changed text.

Corresponding LaTeX code of `./paper-minted.tex`

```
625 This is some text.
626 \change{FL1: text adjusted}{Changed text}.
```

Here is a comment.

Corresponding LaTeX code of `./paper-minted.tex`

```
630 Here is a comment\sidecomment{Comment}.
```

TODO!

Corresponding LaTeX code of `./paper-minted.tex`

```
634 \todo{A lot of text still needs to be produced
    ↪ here}
```

D. Hyphenation

LaTeX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. You might still need to tweak the hyphenation. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also insert `\allowbreak` into the word, as in `application-specific`, but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For that, write “= instead of –, as in application”=specific. This is enabled by an additional configuration of the babel package.

Corresponding L^AT_EX code of ./paper-minted.tex

```
645 In case you write
    ↳ \enquote{application-specific}, then the
    ↳ word will only be hyphenated at the dash.
646 You can also insert \verb1\allowbreak{}1 into
    ↳ the word, as in
    ↳ applica\allowbreak{}tion-specific, but this
    ↳ is much more effort.
647
648 You can now write words containing hyphens
    ↳ which are hyphenated at other places in the
    ↳ word.
649 For that, write \verb1"=1 instead of \verb1-1,
    ↳ as in application"=specific.
650 This is enabled by an additional configuration
    ↳ of the babel package.
```

E. Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: $100 \frac{\text{km}}{\text{h}}$, or by using plain L^AT_EX (and math mode): $100 \frac{\text{km}}{\text{h}}$.

Corresponding L^AT_EX code of ./paper-minted.tex

```
656 Numbers can be written plain text (such as
    ↳ 100), by using the \href{https://ctan.org/}
    ↳ pkg/siunitx}{siunitx} package as follows:
657 \SI{100}{\km\per\hour},
658 or by using plain \LaTeX{} (and math mode):
659 \$100 \frac{\mathit{km}}{h}$.
```

5% of 10 kg

Corresponding L^AT_EX code of ./paper-minted.tex

```
663 \SI{5}{\percent} of \SI{10}{kg}
```

Numbers are automatically grouped: 123 456.

Corresponding L^AT_EX code of ./paper-minted.tex

```
667 Numbers are automatically grouped:
    ↳ \num{123456}.
```

F. Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

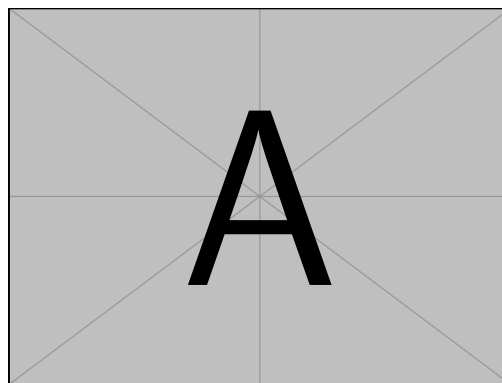


Figure 1: Example figure for zceref demo

Heading1	Heading2
One	Two
Three	Four

Figure 2: Example table for zceref demo

Corresponding L^AT_EX code of ./paper-minted.tex

```
673 Please use the \enquote{enquote command} to
    ↳ quote something.
674 Quoting with "quote" or `quote' also works.
675
```

G. Zref-clever examples

zref-clever demonstration: Zceref (short for zceref[S]) at beginning of sentence, zceref in all other cases.

Figure 1 shows a simple fact, although Figure 1 could also show something else.

Figure 2 shows a simple fact, although Figure 2 could also show something else.

Section III-G shows a simple fact, although Section III-G could also show something else.

Corresponding L^AT_EX code of ./paper-minted.tex

```
705 \Zceref{fig:ex:ceref} shows a simple fact,
    ↳ although \zceref{fig:ex:ceref} could also
    ↳ show something else.
706 \Zceref{tab:ex:ceref} shows a simple fact,
    ↳ although \zceref{tab:ex:ceref} could also
    ↳ show something else.
708 \Zceref{sec:ex:ceref} shows a simple fact,
    ↳ although \zceref{sec:ex:ceref} could also
    ↳ show something else.
```

H. Figures

Figure 3 shows something interesting.



Figure 3: Simple Figure. Based on Scharrer [3].



Figure 4: Ulm Minster. Photo: wkrauth, CC0, via Wikimedia Commons.

Corresponding L^AT_EX code of ./paper-minted.tex

```

715 \Zcref{fig:label} shows something interesting.
716
717 \begin{figure}
718   \centering
719   \includegraphics[width=.8\linewidth]{
    ↪ example-image-golden}
720   \caption[Simple Figure]{
721     Simple Figure.
722     Based on \cite{mwe}.
723   }
724   \label{fig:label}
725 \end{figure}

```

Photos in JPEG or PNG format are included the same way, as shown in Figure 4.

Corresponding L^AT_EX code of ./paper-minted.tex

```

731 \begin{figure}
732   \centering
733   \includegraphics[width=.5\linewidth]{
    ↪ figures/ulm-minster.jpg}
734   \caption{Ulm Minster. Photo: wkrauth, CC0,
    ↪ via Wikimedia Commons.}
735   \label{fig:jpg}

```

```

736 \end{figure}

```

One can span a figure across multiple columns by using `\begin{figure*}`. See Figure 5 as an example.

Corresponding L^AT_EX code of ./paper-minted.tex

```

743 \begin{figure*}
744   \centering
745   % note that \textwidth is used instead of
    ↪ \linewidth
746   % This ensures that the graphics width is 60%
    ↪ of the "page" (text block), and not just
    ↪ 60% of the current text column
747   % See https://tex.stackexchange.com/a/17085/
    ↪ 9075 for details
748   \includegraphics[width=.6\textwidth]{
    ↪ example-image-16x9}
749   \caption{16x9 Figure}
750   \label{fig:16x9}
751 \end{figure*}

```

I. Sub Figures

An example of two sub figures is shown in Figure 6.

Corresponding L^AT_EX code of ./paper-minted.tex

```

760 \begin{figure*}[!t]
761   \centering
762   \subfloat[Case I]{\includegraphics[width=.4
    ↪ \linewidth]{example-image-a}%
    ↪ \label{fig:first_case}}
763   \hfil
764   \subfloat[Case II]{\includegraphics[width=.4
    ↪ \linewidth]{example-image-b}%
    ↪ \label{fig:second_case}}
765   \caption{Example figure with two sub
    ↪ figures.}
766   \label{fig:two_sub_figures}
767   \end{figure*}

```

Note that often IEEE papers with subfigures do not employ subfigure captions (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for `subfig.sty` to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g., `\subfloat[]`. An example is shown in Figure 7.

Corresponding L^AT_EX code of ./paper-minted.tex

```

782 \begin{figure*}[!t]
783   \centering
784   \subfloat[]{\includegraphics[width=.4
    ↪ \linewidth]{example-image-a}%
    ↪ \label{fig:first_case_ieee}}
785   \hfil
786   \subfloat[]{\includegraphics[width=.4
    ↪ \linewidth]{example-image-b}%
    ↪ \label{fig:second_case_ieee}}
787   \caption{}
788   \label{fig:two_sub_figures_ieee}

```

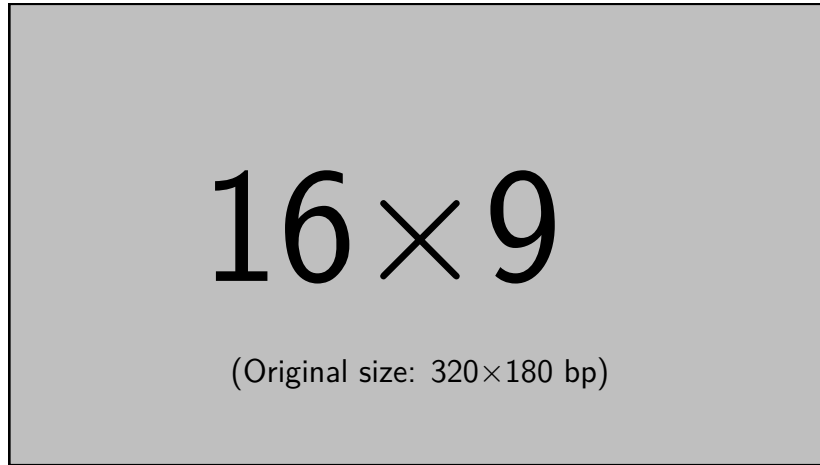
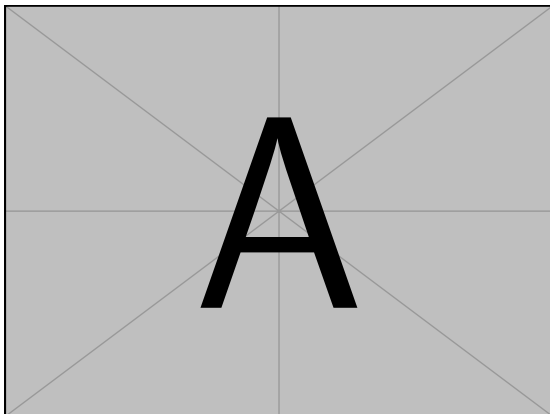
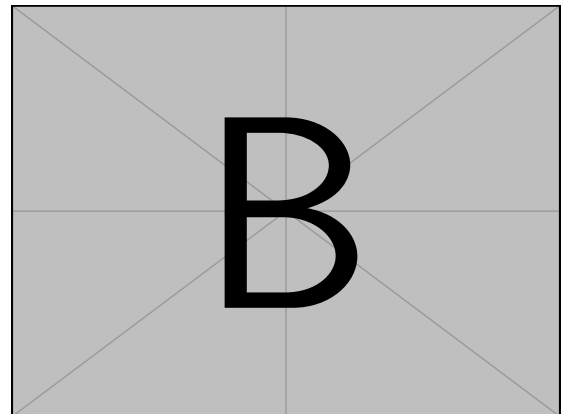


Figure 5: 16x9 Figure

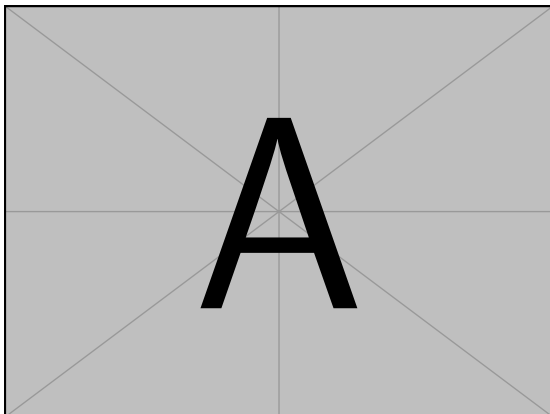


(a) Case I

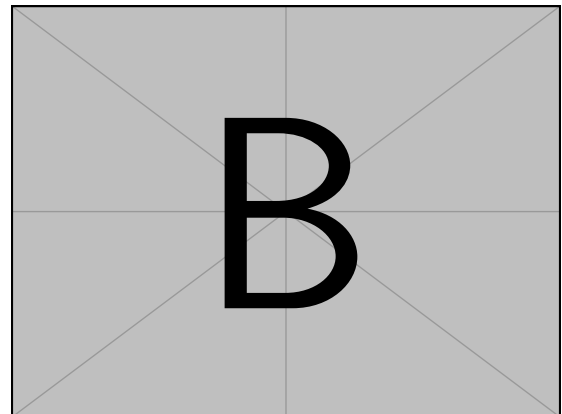


(b) Case II

Figure 6: Example figure with two sub figures.



(a)



(b)

Figure 7: Example figure with two sub figures. IEEE style. (a) The first case. (b) The second case.

```
789 \caption{Example figure with two sub figures.
    ↳ IEEE style. (a) The first case. (b) The
    ↳ second case.}
```

```
790 \label{fig:two_sub_figures_ieee}
791 \end{figure*}
```

J. Figures with tikz

The tikz package creates graphics programmatically. This package generates grids and other regular structures.

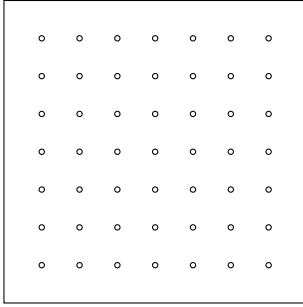


Figure 8: A regular grid generated easily with two for loops.

Corresponding $\text{\LaTeX}$ code of ./paper-minted.tex

```
800 \begin{figure}[ht]
801 \centering
802 \begin{tikzpicture}
803 \draw(0,0) rectangle (4,4);
804 \foreach \x in {0.5,1,1.5,2,2.5,3,3.5}
805 \foreach \y in {0.5,1,1.5,2,2.5,3,3.5}
806 \draw(\x,\y) circle (1pt);
807 \end{tikzpicture}
808 \caption{A regular grid generated easily with
809 \quad \hookrightarrow two for loops.}\label{fig:tikz_example}
809 \end{figure}
```

K. Plots with pgfplots

The pgfplots package plots functions directly in $\text{\LaTeX}$, similar to MATLAB or gnuplot. Some visual examples are available in the package documentation⁵.

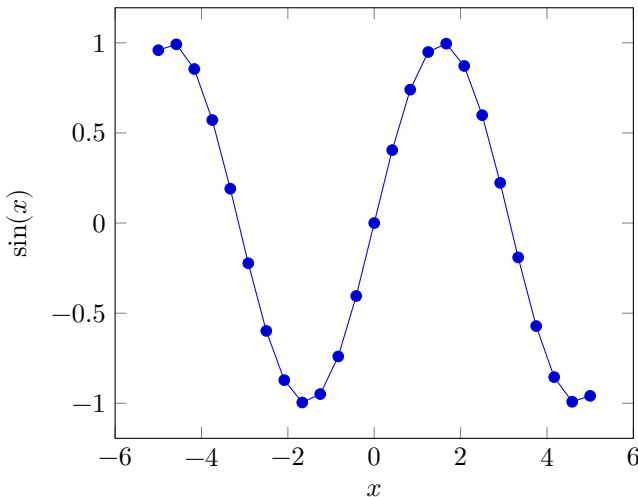


Figure 9: Plot of $\sin(x)$ directly inside the figure environment with pgfplots.

Corresponding $\text{\LaTeX}$ code of ./paper-minted.tex

```
818 \begin{figure}[ht]
819 \centering
820 \begin{tikzpicture}
821 \begin{axis}[xlabel=$x$, ylabel=$\sin(x)$]
822 \addplot {\sin(deg(x))}; % Plot the sine
823 \end{axis}
824 \end{tikzpicture}
825 \caption{Plot of $\sin(x)$ directly inside
826 \quad \hookrightarrow the figure environment with pgfplots.}
827 \label{fig:pgfplots_example}
827 \end{figure}
```

Coordinates can also be given explicitly instead of as a function:

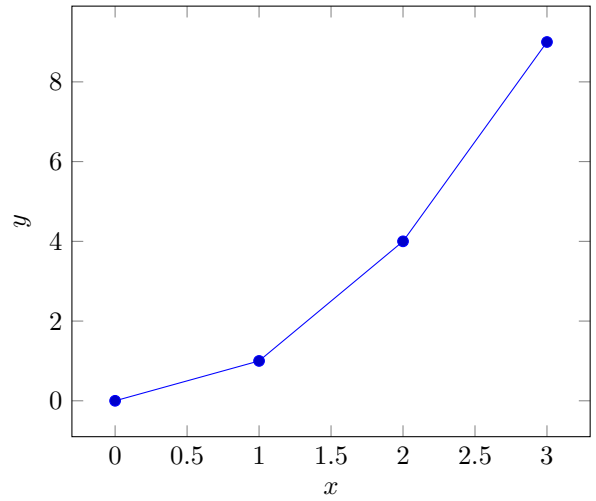


Figure 10: Explicit coordinates plotted with pgfplots.

Corresponding $\text{\LaTeX}$ code of ./paper-minted.tex

```
832 \begin{figure}[ht]
833 \centering
834 \begin{tikzpicture}
835 \begin{axis}[xlabel=$x$, ylabel=$y$]
836 \addplot coordinates {(0,0) (1,1) (2,4)
837 \quad \hookrightarrow (3,9)}; % Plot explicit coordinates
838 \end{axis}
839 \end{tikzpicture}
840 \caption{Explicit coordinates plotted with
841 \quad \hookrightarrow pgfplots.}
842 \label{fig:pgfplots_coors}
842 \end{figure}
```

L. Tables

Note that IEEE does not support $\text{\LaTeX}$ `\begin{table}`, one has to use $\text{\LaTeX}$ `\begin{figure}`.

⁵<http://texdoc.net/pkg/pgfplots>

Figure 11: Simple Table

Heading1	Heading2
One	Two
Three	Four

Figure 12: Table with diagonal line

Row \ Column	A	B
1	x	y

Corresponding L^AT_EX code of ./paper-minted.tex

```

849 \begin{figure}
850   \caption{Simple Table}
851   \label{tab:simple}
852   \centering
853   \begin{tabular}{ll}
854     \toprule
855     Heading1 & Heading2 \\
856     \midrule
857     One      & Two      \\
858     Three    & Four     \\
859     \bottomrule
860   \end{tabular}
861 \end{figure}

```

Corresponding L^AT_EX code of ./paper-minted.tex

```

865 % Adapted from
866 % ↪ https://tex.stackexchange.com/a/468994/9075
867 \begin{figure}
868   \caption{Table with diagonal line}
869   \label{tab:diag}
870   \begin{center}
871     \begin{tabular}{|l|c|c|}
872       \hline
873       \diagbox{Row}{Column} & A & B \\
874       \hline
875       1 & x & y \\
876       \hline
877     \end{tabular}
878   \end{center}
879 \end{figure}

```

M. Source Code

minted is a sophisticated package to enable properly highlighted listings. It uses the pygments library, which in turn requires Python.

Listing 1 shows source code written in XML. line 2 contains a comment.

```

1 <listing name="example">
2   <!-- comment -->
3   <content>not interesting</content>
4 </listing>

```

List. 1: Example XML listing using minted

Corresponding L^AT_EX code of ./paper-minted.tex

```

888 \Zcref{lst:XML} shows source code written in
889 ↪ XML.
890 \refline{line:comment} contains a comment.
891 \begin{listing}[htbp]
892   \begin{
893     ↪ minted}[linenos=true,escapeinside=||]{
894     ↪ xml}
895   <listing name="example">
896     <!-- comment --> |\labelline{line:comment}|
897     <content>not interesting</content>
898   </listing>
899   \end{minted}
900   \caption{Example XML listing using minted}
901   \label{lst:XML}
902 \end{listing}

```

One can also typeset JSON as shown in Listing 2.

```

1 {
2   key: "value"
3 }

```

List. 2: Example JSON listing using minted

Corresponding L^AT_EX code of ./paper-minted.tex

```

906 \begin{listing}[htbp]
907   \begin{
908     ↪ minted}[linenos=true,escapeinside=||]{
909     ↪ json}
910   {
911     key: "value"
912   }
913   \end{minted}
914   \caption{Example JSON listing using minted}
915   \label{lst:fJSON}
916 \end{listing}

```

Java is also possible as shown in Listing 3.

```

1 public class Hello {
2   public static void main (String[] args) {
3     System.out.println("Hello World!");
4   }
5 }

```

List. 3: Java code rendered using minted

Corresponding L^AT_EX code of ./paper-minted.tex

```

920 \begin{listing}[htbp]
921   \begin{
922     ↪ minted}[linenos=true,escapeinside=||]{
923     ↪ java}
924   public class Hello {
925     public static void main (String[] args) {
926       System.out.println("Hello World!");
927     }
928   }
929   \end{minted}

```

```

928 \caption{Java code rendered using minted}
929 \label{lst:flJava}
930 \end{listing}

```

N. Itemization

One can list items as follows:

- Item One
- Item Two

Corresponding L^AT_EX code of ./paper-minted.tex

```

938 \begin{itemize}
939 \item Item One
940 \item Item Two
941 \end{itemize}

```

With the package paralist, one can create itemizations with lesser spacing:

- Item One
- Item Two

Corresponding L^AT_EX code of ./paper-minted.tex

```

947 \begin{compactitem}
948 \item Item One
949 \item Item Two
950 \end{compactitem}

```

One can enumerate items as follows:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of ./paper-minted.tex

```

956 \begin{enumerate}
957 \item Item One
958 \item Item Two
959 \end{enumerate}

```

With the package paralist, one can create enumerations with lesser spacing:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of ./paper-minted.tex

```

965 \begin{compactenum}
966 \item Item One
967 \item Item Two
968 \end{compactenum}

```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

- 1) All these items... 2) ...appear in one line 3) This is enabled by the paralist package.

Corresponding L^AT_EX code of ./paper-minted.tex

```

974 \begin{inparaenum}
975 \item All these items...
976 \item ...appear in one line
977 \item This is enabled by the paralist
    ⇨ package.
978 \end{inparaenum}

```

O. Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding L^AT_EX code of ./paper-minted.tex

```

984 The words \enquote{workflow} and
    ⇨ \enquote{dwarflike} can be copied from the
    ⇨ PDF and pasted to a text file.

```

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).

$\wp(1, 2, 3)$

Corresponding L^AT_EX code of ./paper-minted.tex

```

988 The symbol for powerset is now correct:
    ⇨  $\wpowerset$  and not a Weierstrass p
    ⇨ ( $\wp$ ).
989
990  $\wpowerset(\{1,2,3\})$ 

```

Brackets work as designed: `<test>` One can also input backticks in verbatim text: ``test``.

Corresponding L^AT_EX code of ./paper-minted.tex

```

994 Brackets work as designed:
995 <test>
996 One can also input backticks in verbatim text:
    ⇨ \verb|`test`|.

```

Acronyms can be typeset in running text with `\initialism`, which sets them as small caps that blend into the surrounding lowercase text. It is the lightweight alternative to the full acronym and glossary management (`\gls`, with an automatic list of abbreviations) provided by the thesis variants: `\initialism` only typesets the acronym and does not add it to any list. The macro and the ready-made acronyms (`\OMG`, `\BPEL`, `\BPMN`, `\UML`) are defined in `commands.tex`, where you can add your own.

The UML is maintained by the OMG; related standards include BPEL and BPMN. Any acronym can be set inline with REST or HTTP.

Corresponding L^AT_EX code of `./paper-minted.tex`

```
1002 The \UML is maintained by the \OMG; related
      ↪ standards include \BPEL and \BPMN.
1003 Any acronym can be set inline with
      ↪ \initialism{REST} or \initialism{HTTP}.
```

IV. CONCLUSION AND OUTLOOK

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ACKNOWLEDGMENT

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which comes right before the reference section in your document [4].

REFERENCES

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All links were last followed on October 5, 2020.